DSA06 DATA HANDLING AND VISUALIZATION

Activity: Correlation Detective | Slot C

Case File CD-C22: The Hospital Ward Case

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PART A — ORIGINAL CASE FILE

Briefing

You are the data detective hired by the nursing superintendent of a district hospital. The file holds 18 daily records from the hot weeks, with seven variables. Some relationships can be trusted. Others are traps: a hidden driver, a curve that fools r, and a single wrong record. Find them.

Variables

R name	What it measures
outside_temp	Outside temperature (deg C)
cold_drinks	Canteen cold-drink sales
heat_admits	Heat-related admissions
ward_dev	Ward temperature minus 21 deg C
discomfort	Patient discomfort score
nurses	Nurses on shift
med_errors	Medication errors reported

Evidence: R code (do not edit)

```
set.seed(52856)
n <- 18
outside_temp <- round(rnorm(n, 37.2, 2.7), 1)
cold_drinks <- round(219 + 44.8 * (0.93 * (outside_temp - 37.2) / 2.7 + 0.37 * rnorm(n)),
0)
heat_admits <- round(10 + 2.7 * (0.92 * (outside_temp - 37.2) / 2.7 + 0.29 * rnorm(n)), 0)
ward_dev <- round(seq(-4.1, 4.1, length.out = n)[order(runif(n))], 1)
discomfort <- round(3.3 + 9.4 * (ward_dev^2 + 0.3) + rnorm(n, 0, 0.5), 1)
nurses <- round(rnorm(n, 15, 2.7), 0)
med_errors <- round(4 + 1.3 * (0.08 * (nurses - 15) / 2.7 + 1.00 * rnorm(n)), 0)
nurses[7] <- 24; med_errors[7] <- 9  # row 7 was typed in by a trainee
case <- data.frame(outside_temp, cold_drinks, heat_admits, ward_dev, discomfort, nurses,
med_errors)
```

Tasks

- 1. Use cor(case) to get the 7 × 7 correlation matrix, rounded to 2 decimals. State n and the number of distinct variable pairs. Draw a correlogram. [4]
- 2. Name the strongest positive and strongest negative off-diagonal pairs. Give r and classify each as weak (<0.3), moderate (0.3–0.7), or strong (>0.7). [3]
- 3. Plot discomfort against ward_dev with a straight line, then add a quadratic fit (y ~ x + I(x^2)). Report r and quadratic R-squared. Explain why r misses the relationship. [4]
- 4. Plot med_errors against nurses. Identify the suspicious row. Report r for all rows and without that row, and explain the effect on r. [4]
- 5. Report correlations for cold_drinks with heat_admits, outside_temp with cold_drinks, and outside_temp with heat_admits. Calculate the partial correlation of cold_drinks and heat_admits controlling for outside_temp using residuals. Give a two-sentence causal verdict and one alternative study design. [5]

PART B — WORKED ANSWERS AND R CODE

The scenario provides generated data rather than a printed dataset. Run the code below after creating case to obtain exact numerical correlations and R-squared. This avoids inventing results.

1. Correlation matrix and pairs

There are n = 18 daily records and 7 variables. The number of distinct pairs is 7 × 6 / 2 = 21. The diagonal of the correlation matrix is 1.

```
round(cor(case), 2)

# Correlogram with ggplot2
library(ggplot2)
library(reshape2)
cm <- round(cor(case), 2)
long_cm <- melt(cm, varnames=c("Variable1","Variable2"), value.name="r")
ggplot(long_cm, aes(Variable1, Variable2, fill=r)) +
  geom_tile(color="white") +
  geom_text(aes(label=sprintf("%.2f", r)), size=3) +
  scale_fill_gradient2(low="blue", mid="white", high="red",
                       midpoint=0, limits=c(-1,1)) +
  coord_equal() + theme_minimal() +
  theme(axis.text.x=element_text(angle=45, hjust=1)) +
  labs(title="Hospital Ward Correlation Correlogram", fill="Pearson r",
       x=NULL, y=NULL)
```

Interpretation: values near +1 show strong positive linear association; values near −1 show strong negative linear association; values near 0 show little linear association. Correlation does not prove causation.

2. Strongest positive and negative pairs

```
cm <- cor(case)
diag(cm) <- NA
pairs <- as.data.frame(as.table(cm))
pairs <- pairs[pairs$Var1 < pairs$Var2, ]
pairs <- pairs[order(-abs(pairs$Freq)), ]
head(pairs, 10)
pairs[which.max(pairs$Freq), ]  # strongest positive
pairs[which.min(pairs$Freq), ]  # strongest negative
```

Use the output to report each pair and its r. For this task, classify |r| below 0.30 as weak, 0.30–0.70 as moderate, and above 0.70 as strong.

3. The deceptive zero: discomfort and ward_dev

```
library(ggplot2)
ggplot(case, aes(ward_dev, discomfort)) +
  geom_point() +
  geom_smooth(method="lm", formula=y~x, se=FALSE, color="blue") +
  theme_minimal() + labs(title="Straight-line fit",
    x="Ward temperature deviation (deg C)", y="Discomfort score")

r_linear <- cor(case$ward_dev, case$discomfort)
quad <- lm(discomfort ~ ward_dev + I(ward_dev^2), data=case)
r_linear
summary(quad)$r.squared

ggplot(case, aes(ward_dev, discomfort)) +
  geom_point() +
  stat_smooth(method="lm", formula=y~x+I(x^2), se=FALSE, color="red") +
```

```
  theme_minimal() + labs(title="Quadratic fit",
    x="Ward temperature deviation (deg C)", y="Discomfort score")
```

Interpretation: discomfort was generated mainly from ward_dev squared, creating a U-shaped relationship. Pearson's r measures straight-line association, so the upward and downward sides of a U-shape can cancel and make r close to zero.

4. Outlier alibi: med_errors and nurses

```
plot(case$nurses, case$med_errors, pch=19,
     xlab="Nurses on shift", ylab="Medication errors",
     main="Medication Errors versus Nurses")
text(case$nurses, case$med_errors, labels=seq_len(nrow(case)),
     pos=3, cex=0.7)
r_all <- cor(case$nurses, case$med_errors)
r_without_row7 <- cor(case$nurses[-7], case$med_errors[-7])
r_all
r_without_row7
case[7, ]
```

The supplied code deliberately changes nurses[7] to 24 and med_errors[7] to 9 and labels row 7 as typed in by a trainee. Therefore, row 7 is the intended suspicious record. Compare the two r values to state whether it strengthens, weakens, or changes the direction of the correlation. Document any correction rather than silently deleting data.

5. Prime suspect: partial correlation

```
r_cold_heat <- cor(case$cold_drinks, case$heat_admits)
r_temp_cold <- cor(case$outside_temp, case$cold_drinks)
r_temp_heat <- cor(case$outside_temp, case$heat_admits)
r_cold_heat
r_temp_cold
r_temp_heat

res_cold <- resid(lm(cold_drinks ~ outside_temp, data=case))
res_heat <- resid(lm(heat_admits ~ outside_temp, data=case))
partial_r <- cor(res_cold, res_heat)
partial_r
```

Verdict: cold-drink sales and heat admissions may be correlated because both tend to rise with outside temperature; this does not prove cold-drink sales cause admissions. A randomized intervention could test causation if ethical and practical, while a natural experiment or longitudinal study controlling for weather and other confounders may be an alternative.

Quick conclusion

- The dataset contains 18 records, 7 variables, and 21 distinct variable pairs.
- A correlation matrix summarizes linear relationships but can miss curved patterns.
- The discomfort relationship is quadratic, so a quadratic model is more appropriate than a straight line.
- Row 7 is deliberately altered and should be checked as a possible data-entry error.
- Partial correlation tests whether cold-drink sales and admissions remain associated after accounting for outside temperature; it still does not prove causation.